MDENet: Multidomain Differential Excavating Network for Remote Sensing Image Change Detection

Jinyang Liu, Shutao Li, *Fellow, IEEE*, Renwei Dian, *Member, IEEE*, Ze Song, and Xudong Kang, *Senior Member, IEEE*

***Abstract*— Remote sensing image change detection can analyze alterations on the Earth's surface within a specific region. However, the accuracy of change detection has consistently been hindered by the style differences in captured images caused by seasonal or lighting variations, as well as the challenge of distinguishing similar features between the background and foreground in the scene. To this end, a multidomain differential excavating network (MDENet) for change detection is introduced. Using the novel multidomain differential collaboration module (MDCM) to precisely capture object features on the frequency and spatial domains across diverse temporal domains, it enables simultaneous querying of global and local change information. Moreover, the multineighborhood frequency gate attention (MFGatt) is devised to eliminate the impact of image style relevance information and consolidate attention toward object localization, thereby enhancing the adaptability of the network to variations in image style. Extensive experiments have illustrated that our proposed network achieves better detection accuracy compared with current state-of-the-art (SOTA) methods on various datasets.**

***Index Terms*— Deep learning, frequency-domain analysis, remote sensing image change detection.**

I. Introduction

TO ACCURATELY detect surface changes, such as cultivated land change, urban expansion, and natural disasters, a large number of remote sensing change detection (RSCD) methods have been used [1], [2], [3]. These methods extract surface change information by analyzing remote sensing data from the same region at different time nodes [4], [5]. With the development of RSCD, more and more high-quality methods are being proposed [6], [7], [8], [9], [10]. Due to the unpredictability of lighting conditions, facility modifications, and periodic changes in the environment during imaging, the accuracy of these methods is still constrained by two major challenges: the style differences between images obtained at different time points make it challenging to identify features of the same type, and the presence of similar features in the background and the detected objects leads to interference in the detection results.

To be more specific, traditional change detection methods are limited by their single feature extraction method, making it difficult to adapt to different styles of detected images. In addition, its differential information detection operator is severely constrained by empirical design, resulting in poor discernibility of background and foreground features. To overcome these limitations, deep-learning-based change detection methods are widely used [11], [12], [13], [14]. They typically use multiple branches to extract features from source images captured at different times, which alleviates the impact of style differences between images to a certain extent [15]. However, convolutional layers only focus on local information and lead to the generated detection model struggling to achieve global feature analysis [16]. Currently, transformer-based change detection methods have been specifically proposed to establish global dependencies of features over time or space by querying the feature correlations between different regions of the source images [17]. These methods are accompanied by a significant amount of attention mechanisms, which results in a considerable computational burden.

In response to the aforementioned challenges, a multidomain differential excavating network (MDENet) is used for change detection, as shown in Fig. 1. To enhance the accuracy of the model in discriminating background and foreground features, a new multidomain differential collaboration module (MDCM) is proposed. It gradually uses the complementary properties from the shallow spatial domain, the deep spatial domain, and the frequency domain to synthesize more accurate changing features. Among them, the shallow spatial-domain change feature is used to compensate for the ability to perceive detailed information of the deep spatial-domain change feature on changing objects. In addition, profiting from the global feature mining capability in the frequency domain [18], [19], [20], the MDCM generates change feature maps from both local and global perspectives by combining spatial- and frequency-domain information. A multineighborhood frequency gate attention (MFGatt) is designed to address the problem of image style differences. It uses a gated selector

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Jinyang Liu, Shutao Li, and Ze Song are with the College of Electrical and Information Engineering and the Key Laboratory of Visual Perception and Artificial Intelligence of Hunan Province, Hunan University, Changsha 410082, China (e-mail: jinyangliu@hnu.edu.cn; shutao_li@hnu.edu.cn; songz@hnu.edu.cn).

Renwei Dian and Xudong Kang are with the School of Robotics and the Key Laboratory of Visual Perception and Artificial Intelligence of Hunan Province, Hunan University, Changsha 410082, China (e-mail: drw@hnu.edu.cn; xudong_kang@163.com).

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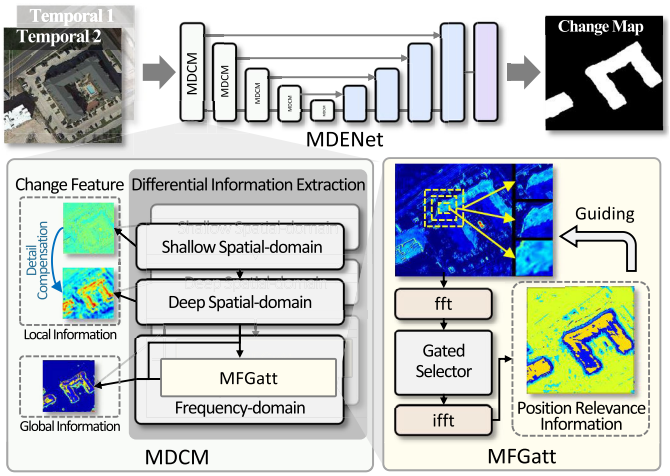


Fig. 1. Thumbnail of MDENet. It uses the proposed MDCM to extract different types of change features in different domains. The focus of the network is guided to the position of the change object using MFGatt.

in the frequency domain to separate image style relevance information, while using the retained object position relevance information to guide the extracted change features from different neighborhoods [21], [22]. In this way, the interest of the model is unified to the position of changing objects, guiding the generation of feature difference maps in the frequency domain.

The main contributions are summarized as follows.

1) We propose a new MDCM for constructing the MDENet, which distinguishes foreground and background features by analyzing the differences in global and local information extracted from different domains.
2) An MFGatt is designed to adapt to the style differences of images and enhance the sensitivity of the model to the position of changing objects.
3) Plentiful experiments have shown that the proposed MDENet can achieve more accurate change detection on different datasets compared with the state-of-the-art (SOTA) methods.

II. Related Works

With the development of RSCD, mainstream methods are focused on using deep learning for implementation. Therefore, we emphatically introduce the methods of RSCD based on deep learning. In addition, frequency-domain analysis has also been widely used in RSCD, and the RSCD methods based on frequency-domain analysis are discussed in detail.

A. Deep-Learning-Based RSCD

The deep-learning-based RSCD method has strong feature representation ability and can adapt to large-scale data processing tasks. Various methods based on convolutional neural networks (CNNs) have been proposed [23], [24], [25], [26], [27], [28]. Among them, dual-branch multilevel intertemporal network (DMINET) improves the interaction degree between multiscale features of different images by designing the intertemporal joint-attention (JointAtt) module [29]. To enhance the sensitivity of the detection model to edge features, region detail preserving network (RDP-Net) proposes an edge loss to constrain the ability of the network to distinguish detailed information [30]. Feature hierarchical differentiation (FHD) introduces global and local attention to fuse the features of different inputs to establish temporal relationships between images [31]. In addition, more and more lightweight networks are being proposed to achieve efficient RSCD. In detail, ultralightweight spatial-spectral feature cooperation (USSFC) has proposed a multiscale decoupled convolution for low computational cost multiscale feature extraction [32]. A2Net aggregates features of different scales and calculates feature differences before sending them to an iterative feature extraction module to extract change information [33]. LightCDNet has designed a lightweight change detection network from the perspective of preserving the integrity of input image information [34].

With the development of deep learning, transformers have been widely applied to various image processing tasks [35], [36], [37], [38], [39]. ChangeFormer implements multiscale feature encoding of different images through the transformer and generates detection results using a lightweight multi-layer perception (MLP) decoder [37]. Cross-temporal context learning network (CCLNet) extracts semantic features of images through a lightweight backbone network and designs a new time-domain context aware transformer to enhance feature correlation between different images [17].

B. Frequency-Domain Analysis for RSCD

Some surface features may exhibit more significant changes in the frequency domain, and the analysis based on frequency-domain signals is more robust to the noise data, making it widely used in RSCD. To improve the ability of the model to extract edge information from the detected object, HFA-Net proposes a high-frequency attention mechanism, which realizes the localization of the detection object and increases the accuracy of detection by enhancing high-frequency information [40]. Cheng et al. [41] design a Laplacian pyramid expansion module (LPEM) to ensure the integrity of high-frequency information during the decoding process and achieve accurate construction of changing features. Enhanced self-attention network for remote sensing building change detection (ESACD) uses a self-attention mechanism to construct two types of attention modules for low-frequency and high-frequency information and then fuses the extracted features, aiming to improve the representation ability of the model [42]. Similarly, a dual cross attention block (DCAB) is used in dual cross-attention-based transformer (DCAT), which combines transformer and convolution operations to extract high-frequency and low-frequency information from different images, respectively [43]. To improve the accuracy of change detection, Gao et al. [44] first separate the significant regions based on frequency-domain analysis and achieve more precise classification through random multigraphs. In unsupervised change detection algorithms, ASGF generates complementary features in both spatial and frequency domains to learn pseudolabels generated by clustering techniques [45].

The methods mentioned above mostly adopt a way of processing high- and low-frequency information separately.

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However, they often overlook the correlation between low- and high-frequency information and do not fully use the global information that can be extracted from the frequency domain. Significantly, many existing detection models use CNN for feature extraction in the spatial domain, but they are limited by the local feature extraction nature of convolution operations, making it difficult to capture global information effectively [32].

Our proposed MDCM leverages the global information extracted from the frequency domain to complement the local information extracted from the spatial domain, which achieves an effective combination of the frequency domain and spatial domain. In addition, the differences in unchanged areas of source images are often caused by style differences between the images, and frequency-domain analysis can effectively separate image style relevance information from object position relevance information [46], [47]. Inspired by this, we design an MFGatt that selects position relevance information in the frequency domain to guide the network, which alleviates the impact of image style differences on the detection task.

III. Methodology

In this section, we provide a detailed description of the two important parts involved in MDENet, namely, the MDCM and the MFGatt. To give a more comprehensive description of the proposed detection method, we also provide an overview of the MDENet. In addition, MDENet-s has also been mentioned as a low parameter version of MDENet.

A. Multidomain Differential Collaboration Module

To address the issue of background and foreground features being easily confused during detection, we propose a multidomain difference cooperation module called MDCM, and its specific structure is shown in Fig. 2. As features in the frequency-domain possess global characteristics, MDCM establishes the relationship of feature differences between spatial and frequency domains, achieving simultaneous analysis of local and global information [18], [19], [20]. Furthermore, to enhance the perception of the detection model to the details in the objects, we adopt a module that combines changes in shallow spatial and deep spatial features. It simultaneously strengthens the semantic information mining ability of the network to differentiate the features between detected objects and other types of objects.

In detail, MDCM mainly includes units for extracting different information in the shallow spatial-domain, the deep spatial domain, and the frequency domain differential information extraction unit. The change feature ϕ_{mdc} generated by it can be represented as follows:

$$\begin{aligned}\phi_{\mathrm{mdc}} &= \mathbf{W}_{\mathrm{cp}}\left(\phi_{Vss}, \phi_{Vds}, \phi_{Vf}\right)\\ \phi_{V\mathrm{ss}} &= \phi_s^1 - \phi_s^2\\ \phi_{V\mathrm{ds}} &= \phi_d^1 - \phi_d^2\\ \phi_{Vf} &= \phi_f^1 - \phi_f^2\end{aligned} \tag{1}$$

where $\mathbf{W}_{\mathrm{cp}}$ represents merging the input elements and using a 3×3 convolution to extract feature. $\phi_{V\mathrm{ss}}$, $\phi_{V\mathrm{ds}}$, and ϕ_{Vf}, respectively, refer to the shallow spatial-domain change feature, deep spatial-domain change feature, and frequency-domain change feature, which are obtained by subtracting different types of features produced by images at different time nodes.

SDIE_s and SDIE_d are the units for extracting spatial-domain difference information in shallow and deep layers, respectively. They are composed of two concatenated 3×3 convolutions, which are set to obtain local information

$$\phi_s^1 = \mathrm{SDIE}_s(\phi^1), \quad \phi_s^2 = \mathrm{SDIE}_s(\phi^2) \tag{2}$$

where ϕ_s^1 and ϕ_s^2 represent shallow spatial features generated by the prechange image feature ϕ^1 and the postchange image feature ϕ^2, respectively. ϕ_d^1, ϕ_d^2 are deep spatial features that are generated based on shallow spatial features and can be represented as

$$\phi_d^1 = \mathrm{SDIE}_d\left(\phi_s^1\right), \quad \phi_d^2 = \mathrm{SDIE}_d\left(\phi_s^2\right). \tag{3}$$

Furthermore, the features in the frequency domain are obtained based on the deep spatial features and represented as

$$\phi_f^1 = \mathrm{FDIE}\left(\phi_d^1\right), \quad \phi_f^2 = \mathrm{FDIE}\left(\phi_d^2\right) \tag{4}$$

where FDIE is the frequency-domain dimension difference information extraction unit. To maintain the efficiency of feature propagation, a 1×1 convolution is selected to establish the connection between the input and the output of FDIE, which is used in parallel with MFGatt. Given an input tensor ϕ, FDIE can be described as

$$\mathrm{FDIE}(\phi) = \mathbf{W}_c\left(\mathbf{W}_q(\phi), \mathrm{MFGatt}(\phi)\right) \tag{5}$$

where $\mathbf{W}_c$, $\mathbf{W}_q$, and MFGatt, respectively, refer to concatenation operation, 1×1 convolution, and multineighborhood frequency gate attention.

The deep spatial features ϕ_d^1 and ϕ_d^2 generated by MDCM are used as inputs for the next scale of MDCM after the max-pooling operator (MaxP), and the corresponding outputs are denoted as ϕ_m^1 and ϕ_m^2. Thus, MDCM can be represented as

$$\begin{aligned}\phi_m^1 &= \mathrm{MaxP}\left(\phi_d^1\right)\\ \phi_m^2 &= \mathrm{MaxP}\left(\phi_d^2\right)\\ \phi_{\mathrm{mdc}}, \phi_m^1, \phi_m^2 &= \mathrm{MDCM}\left(\phi^1, \phi^2\right).\end{aligned} \tag{6}$$

Significantly, to ensure that the output features are consistent with the scale of the previous change feature, all the features before the concatenation operation are upsampled, which are used for processing minimum scale features of the MDCM.

B. Multineighborhood Frequency Gate Attention

Images of the same scene captured at different times often exhibit significant style differences, causing notable variations in features of the same type within the region of interest and disrupting the accuracy of detection. To address this challenge, we propose a multi MFGatt mechanism, as illustrated in Fig. 3. It guides the model to determine the object position relevance information in the image through frequency-domain analysis,

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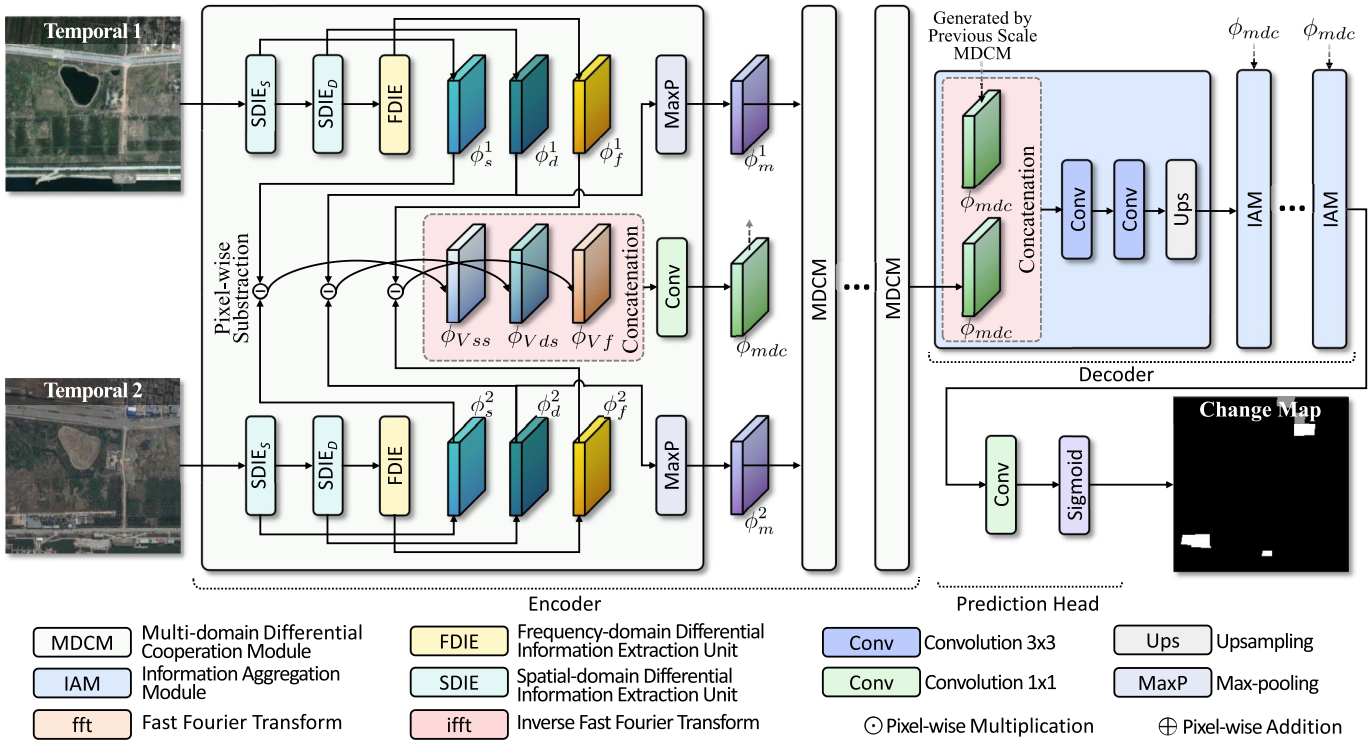


Fig. 2. Pipeline of the proposed MDENet. It consists of three main parts: encoding, decoding, and prediction head. MDCM and IAM play roles in feature extraction, aggregation, and reconstruction during the encoding and decoding process.

and then compares and extracts variation features, eliminating the interference of image style relevance information.

To ensure the integrity of the extracted features of MFGatt, we first use 3×3 atrous convolutions $\mathbf{W}_{dx}$ with three different atrous rates $x \in [1, 3, 5]$ for feature exploration. By representing the input as ϕ, the above process can be expressed as follows:

$$\phi_d = \mathbf{W}_{d1}(\phi) + \mathbf{W}_{d3}(\phi) + \mathbf{W}_{d5}(\phi). \tag{7}$$

Then, the generated feature ϕ_d is transformed into the frequency domain through the fast Fourier transform, and the image style relevance information is filtered out using a gated selector.

Finally, the attention map in the frequency domain is generated through the Sigmoid function, guiding the model to adaptively partition the parts of the features that are more relevant to the objects. Therefore, MFGatt can be written as

$$\text{MFGatt}(\phi_d) = \phi_d \odot \text{Sigmoid}(\mathbf{F}_i(\text{Gate}(\mathbf{F}(\phi_d)))) \tag{8}$$

where $\mathbf{F}$ and $\mathbf{F}_i$ represent the fast Fourier transform and the inverse transform, respectively. $\odot$ is the pixelwise multiplication operation. Sigmoid is used to constrain the interval of the output value between 0 and 1.

The Gate represents the gated selector. It designs a self-query process to filter information, which can be specifically represented as

$$\text{Gate}(\phi_{df}) = \mathbf{W}_p(\phi_{df}) \odot \text{GeLU}(\mathbf{W}_p(\phi_{df})) \tag{9}$$

where $\mathbf{W}_p$ refers to 3×3 convolution, and Gaussian error linear unit (GeLU) is the activation function. ϕ_{df} represents the feature after the fast Fourier transform.

C. Overview of MDENet

The proposed MDENet is a U-shaped change detection network, and its network structure is shown in Fig. 2. First, it achieves feature encoding at different scales by simultaneously obtaining information from different images through the MDCM, generating change features ϕ_{mdc}. During the feature decoding process, the information aggregation module (IAM) is used, which consists of a transposed convolution for upsampling (Ups) and two 3×3 convolutions in series connection to remove redundant information from the change feature ϕ_{mdc} of different scales, which realizes the aggregation and reconstruction of change features. The IAM used for processing large-scale features does not include upsampling operations, and it can be described as

$$\text{IAM}(\phi_{\text{mdc}}) = \text{Ups}\left(\mathbf{W}_p\left(\mathbf{W}_{\text{cp}}\left(\phi_{\text{mdc}}, \phi'_{\text{mdc}}\right)\right)\right) \tag{10}$$

where ϕ'_{mdc} refers to the change feature of the previous level.

Finally, the decoded features generate a change map through a simple prediction head. Denoting the prediction head and its input as head and ϕ, respectively. The head can be described as

$$\text{Head}(\phi) = \text{Sigmoid}(\mathbf{W}_q(\phi)). \tag{11}$$

The binary cross-entropy loss function is used to measure the difference between the ground truth and the change map, which can constrain the training process of the network. Thus, the loss function can be described as

$$\mathcal{L}_{\text{bce}} = -\frac{1}{N_b}\sum_i g_i \log(p_i) \tag{12}$$

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where N_b represents the total number of samples during the training process, and g_i and p_i represent the ground truth of the ith sample and the corresponding change map, respectively.

D. MDENet-s

In adapting to compute platforms with lower computing power and meeting the needs of algorithm deployment, we propose a low parameter version of MDENet (MDENet-s). It reduces the number of output channels in all the convolutional layers to a quarter of the original without changing the MDENet structure, which significantly reduces the number of model parameters. MDENet-s is capable of achieving higher computational efficiency while maintaining a similar level of change detection accuracy to the original MDENet, reflecting the effectiveness of the designed network structure for the change detection task.

IV. Experimental Results and Discussion

In this section, the relevant experiments are introduced in detail. First, we illustrate the implementation methods, datasets, evaluation indicators, and other environmental settings related to the experiment. Second, the proposed method is compared with the advanced methods. Third, we specifically analyze the computational efficiency of different change detection models. Finally, ablation experiments are conducted to demonstrate the effectiveness of the proposed modules.

A. Experimental Setting

The proposed network is deployed on NVIDIA Quadro RTX 6000 GPU and 2.10-GHz Intel Xeon Silver 4216 CPU, using Python as the framework for code writing. During training, we use adaptive moment estimation (ADAM) as the optimizer, with the momentum set to 0.99 and the weight decay value of 0.0005. The learning rate is set to 0.0001, and the overall training lasted for 200 epochs.

To accurately reflect the effectiveness of the proposed change detection network, we select three publicly available datasets for validation.

LEVIR-CD[1] is a high-resolution dataset focused on detecting changes in architectural scenes in remote sensing [48]. It consists of 637 pairs of 1024 × 1024 images from different time points. To facilitate data retrieval and processing, we split it into pairs of 256 × 256 images, with 7120/1024/2048 pairs used for training/validation/testing, respectively.

WHU-CD[2] is also a building CD dataset illustrating changes in the same region between different years, with a resolution of 32 507 × 15 345 for the original image [49]. Similarly, it has been cropped into pairs of 256 × 256 images to fit the training equipment, with 6096/762/762 pairs corresponding to training/validation/testing data.

DSIFN-CD[3] contains change scenes from six different cities, collected from Google Earth [50]. The dataset has a resolution of 512 × 512. Specifically, 3600/340/48 image pairs are, respectively, used for training/validation/testing.

[1]https://chenhao.in/LEVIR/

[2]https://study.rsgis.whu.edu.cn/pages/download/building dataset.html

[3]https://github.com/GeoZcx/A-deeply-supervised-image-fusion-network-for-change-detection-in-remote-sensing-images/tree/master/dataset

In terms of quantitative testing in the experiment, five different indicators have been selected, namely, precision (Pre), recall (Rec), F1-score (F1), intersection over union (IoU), and overall accuracy (OA). The larger the value of the above indicators, the better the detection performance. They can be defined as follows:

$$\begin{aligned} \text{Pre} &= \frac{\text{TP}}{\text{TP}+\text{FP}} \\ \text{Rec} &= \frac{\text{TP}}{\text{TP}+\text{FN}} \\ \text{F1} &= \frac{2}{\text{Pre}^{-1}+\text{Rec}^{-1}} \\ \text{IoU} &= \frac{\text{TP}}{\text{TP}+\text{FN}+\text{FP}} \\ \text{OA} &= \frac{\text{TP}+\text{FN}}{\text{TP}+\text{TN}+\text{FN}+\text{FP}} \end{aligned} \tag{13}$$

where TP, TN, FP, and FN represent true positive, true negative, false positive, and false negative, respectively.

B. Comparison to the State-of-the-Art

In all, 12 SOTA methods are selected for comparison with the proposed method: full convolutional early fusion (FC-EF) [6], fully convolutional siamese - difference (FC-Siam-Di) [6], fully convolutional siamese - concatenation (FC-Siam-Conc) [6], spatial–temporal attention-based network (STANet) [48], image fusion network (IFNet) [50], densely connected siamese network for change detection (SNUNet) [24], improved separability deep learning network (ISNet) [51], USSFC [32], DMINET [29], CCLNet [17], global-aware siamese network (GASNET) [52], and cross-level attentive feature aggregation (CLAFA) [53]. Among them, USSFC and DMINET are built by lightweight networks. The results of all these methods are obtained based on published precision or retrained by open-source code. Each method is described in detail as follows.

1) **FC-EF** is one of the early representative change detection methods, which is based on fully convolutional neural network (FCN) to achieve change detection.
2) **FC-Siam-Di** has constructed a detection method for multilevel feature extraction and fusion, achieving differential information extraction of images at different time nodes.
3) **FC-Siam-Conc** has a similar change detection method to FC-Siam-Di, which uses convolution to extract features from images at different time nodes.
4) **STANet** emphasizes the spatiotemporal relationship between the detected images to obtain more accurate change detection results.
5) **IFNet** focuses on analyzing the variations between different images in scale, implementing change analysis of input images.
6) **SNUNet** combines dense linking methods and uses channel attention to enhance the feature representation capability of the network.
7) **ISNet** uses channel attention in the backbone networks to enhance the ability to extract semantic information and uses spatial attention to obtain more accurate object location information.

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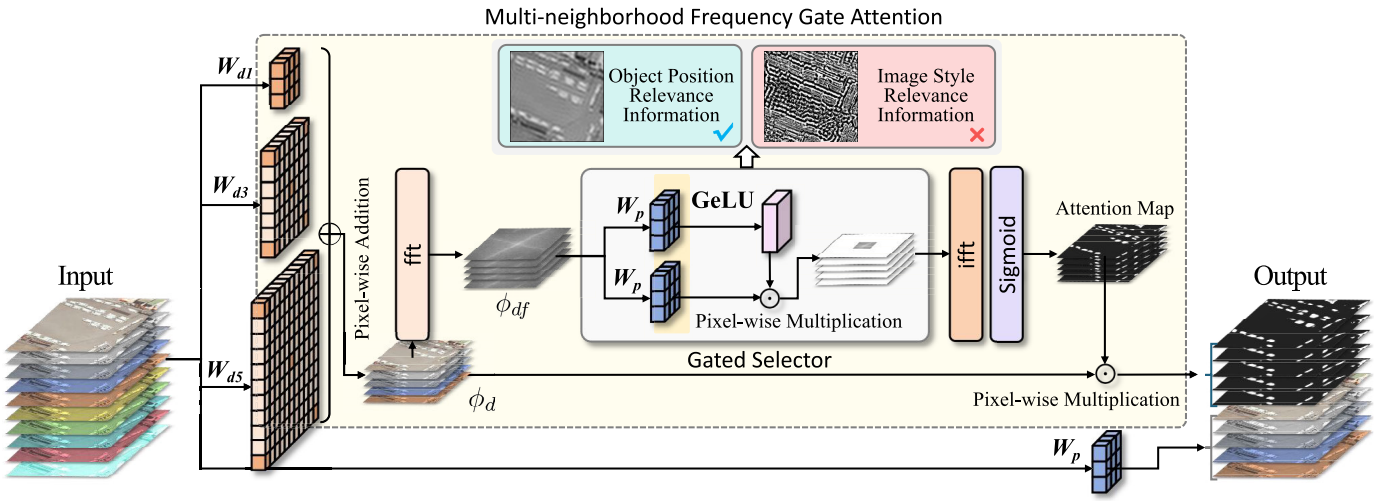


Fig. 3. Illustration of FDIE and the composition of MFGatt. Both input and output represent the corresponding tensors. The extracted input features are transformed into the frequency domain and filtered through a gated selector to preserve the object position relevance information. This information is used to redirect the attention of the change detection model to the detected object, reducing the impact of image style on detection accuracy.

TABLE I

COMPARISON OF CHANGE DETECTION PERFORMANCE ON DIFFERENT DATASETS. THE SUBSCRIPT REPRESENTS THE TIME OF PUBLICATION. MDENET-S IS THE LOW PARAMETER VERSION OF MDENET (RED: THE BEST)

Method	LEVIR-CD					WHU-CD					DSIFN-CD				
	Pre	Rec	F1	IoU	OA	Pre	Rec	F1	IoU	OA	Pre	Rec	F1	IoU	OA
FC-EF$_{18}$	86.91	80.17	83.40	71.53	98.39	71.63	67.25	69.37	53.11	97.61	72.61	52.73	61.09	43.98	88.59
FC-Siam-Di$_{18}$	89.53	83.31	86.31	75.92	98.67	47.33	77.66	58.81	41.66	95.63	59.67	65.71	62.54	45.50	86.63
FC-Siam-Conc$_{18}$	91.99	76.77	83.69	71.96	98.49	60.88	73.58	66.63	49.95	97.04	66.45	54.21	59.71	42.56	87.57
STANet$_{20}$	83.81	91.00	87.26	77.40	98.66	79.37	85.50	82.32	69.95	97.42	67.71	61.68	64.56	47.66	88.49
IFNet$_{20}$	94.02	82.93	88.13	78.77	98.87	96.91	73.19	83.40	71.52	98.52	67.86	53.94	60.10	42.96	87.83
SNUNet$_{22}$	89.18	87.17	88.16	78.83	98.82	85.60	81.49	83.50	71.67	98.83	60.60	72.89	66.18	49.45	87.34
ISNet$_{22}$	89.22	89.02	89.12	80.37	98.89	74.06	93.48	82.64	70.42	98.46	39.03	83.27	53.15	36.19	76.50
USSFC$_{23}$	88.46	91.80	90.10	81.98	99.01	92.43	93.56	92.99	86.90	99.42	71.38	58.14	64.09	47.15	86.41
DMINET$_{23}$	92.52	89.95	90.71	82.99	99.07	93.84	86.25	88.69	79.68	98.97	51.29	73.45	60.40	43.27	84.58
CCLNet$_{23}$	91.15	89.85	90.49	82.64	99.04	86.92	89.37	88.13	78.78	99.00	65.62	55.46	60.11	42.97	88.22
GASNET$_{23}$	87.79	93.34	90.48	82.62	99.00	90.16	92.72	91.42	84.20	99.28	52.34	72.80	60.90	43.78	84.12
CLAFA$_{23}$	92.72	87.63	90.10	81.99	99.02	92.12	94.50	93.30	87.43	99.44	54.82	66.53	60.11	42.97	85.87
MDENet-s	87.87	92.48	90.12	82.01	99.02	91.38	92.74	92.06	85.28	99.35	58.39	75.98	66.04	49.29	89.79
MDENet	89.55	92.68	91.09	83.63	99.11	91.54	95.49	93.47	87.74	99.47	64.61	75.66	69.70	53.49	90.45

8) **USSFC** has designed a multiscale decoupled convolution to achieve rapid capture of changing features.
9) **DMINET** encodes features of different scales by combining self-attention and cross-attention.
10) **CCLNet** is a transformer-based change detection model that achieves detection by mining temporal contextual changing features.
11) **GASNET** focuses on analyzing the relationship between global and foreground features of images, achieving context information mining of images.
12) **CLAFA** deeply considers how to achieve an accurate representation of semantic information through sufficient interaction of features at different levels.

The quantitative experimental results on the LEVIR-CD dataset are shown in Table I. Our proposed MDENet achieves better performance in indicators F1, IoU, and OA compared with other advanced methods and also achieves second place in Rec. In addition, compared with detection methods such as CCLNet and GASNET, which concentrate on learning image context association information, the proposed method has more prominent detection accuracy, reflecting its stronger global information acquisition ability.

Due to the large time span between the prechange and postchange images in the WHU-CD dataset, there are significant image style differences. Table I displays the detection results on this dataset. It is easy to note from Table I that the proposed method has achieved varying degrees of leadership in indicators Rec, F1, IoU, and OA compared with the second ranked method. Compared with other SOTA methods, MDENet-s also has a good ranking in these indicators, which reflects that the proposed low parameter version of the model can achieve accurate and faster change detection.

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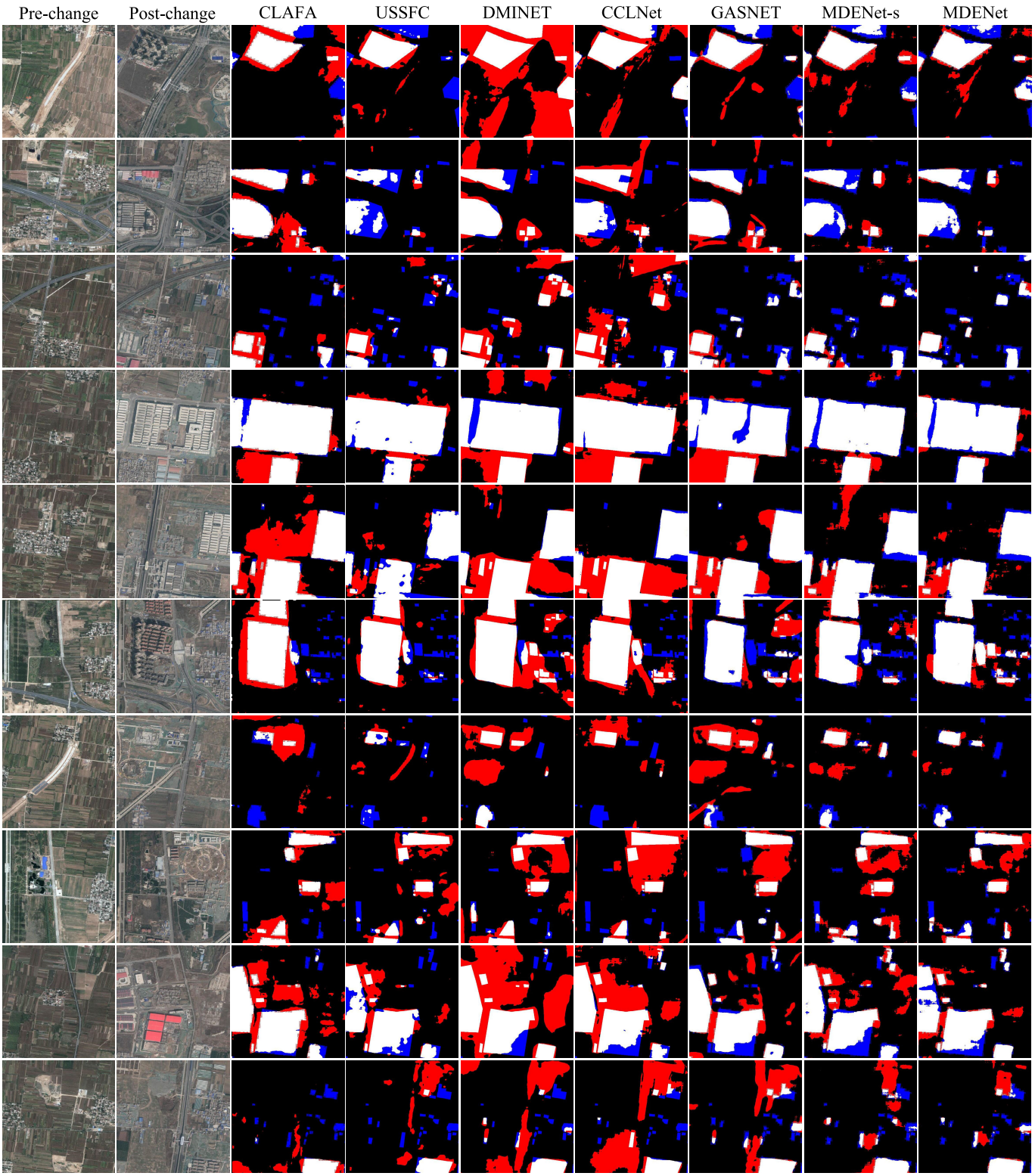


Fig. 4. Qualitative experiments on the DSIFN-CD dataset. The red, blue, black, and white areas in the figure represent false positive, false negative, true negative, and true positive, respectively.

The compared models generally exhibit lower detection accuracy on the DSIFN-CD dataset compared with the LEVIR-CD and WHU-CD datasets. This situation might be caused by the less pronounced land cover change features in the DSIFN-CD dataset, which proves that it can better reflect the recognition ability of the change detection model for background and foreground. In Table I, the proposed method achieves a respective increase of 3.52%, 4.04%, and 1.86% in F1, IoU, and OA, respectively, compared with the SOTA method with the highest value. It is noteworthy

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TABLE II
COMPARISON OF COMPUTATIONAL EFFICIENCY WITH SOTA METHODS. MDENET-S IS THE LOW PARAMETER VERSION OF MDENET (RED: THE BEST)

	USSFC	DMINET	CCLNet	GASNET	CLAFA	**MDENet**	**MDENet-s**
FLOPs(G)	9.7203	29.1029	52.0073	579.9309	16.5972	108.5863	6.9636
Params(M)	1.5190	6.2422	11.3339	23.5217	5.0583	49.2970	3.0919

that MDENet-s ranks first overall in Rec, and it achieves either first or second place in other indicators compared with other advanced methods. Furthermore, qualitative results are depicted in Fig. 4. Upon observation, it is evident that the proposed method exhibits significantly fewer false positives (depicted in red) and false negatives (depicted in blue) in comparison to the visual results of other models, indicating that MDENet demonstrates enhanced analysis capabilities for foreground and background features.

C. Computational Efficiency Analysis

Computational efficiency is an important indicator for evaluating the quality of a change detection model. In the experiment, we select floating-point operations per second (FLOPs) and training parameters as measurement indicators, and their specific results are shown in Table II. The resolution size of the testing data is 256 × 256. USSFC and DMINET are two lightweight detection methods, but they are both lower than MDENet-s in FLOPs. Compared with other methods, the number of parameters in MDENet-s is also similar to the USSFC. In addition, when conducting comparative experiments without including the original version, MDENet-s ranks second in most indicators on the dataset and can even rank first in some cases. Therefore, by comprehensively considering comparative experiments and efficiency analysis experiments, it can be proven that the network framework of MDENet can achieve fast and accurate change detection.

D. Ablation Experiments

The proposed MDENet consists of two significant components: the MDCM and the MFGatt. These components are designed to address the challenges posed by image style differences and similarities in background and foreground features. To further illustrate the impact of these two components, we conduct relevant ablation experiments as follows.

1) *Analysis of MDCM:* To accurately reflect the role of shallow spatial-domain, deep spatial-domain, and frequency-domain differential information extraction unit in the MDCM, SDIE$_s$, SDIE$_d$, and FDIE are removed or selected. We conduct ablation experiments on three datasets, and the specific experimental results are shown in Table III. When removing the extraction units of the deep spatial-domain differential information extraction unit, almost every indicator shows a decrease in each dataset, which indicates that the global analysis ability of the frequency-domain differential information extraction unit is particularly important. Overall, the three key indicators F1, IoU, and OA demonstrate that our proposed method of parallel extracting the global and local information in frequency and spatial domains has better detection performance.

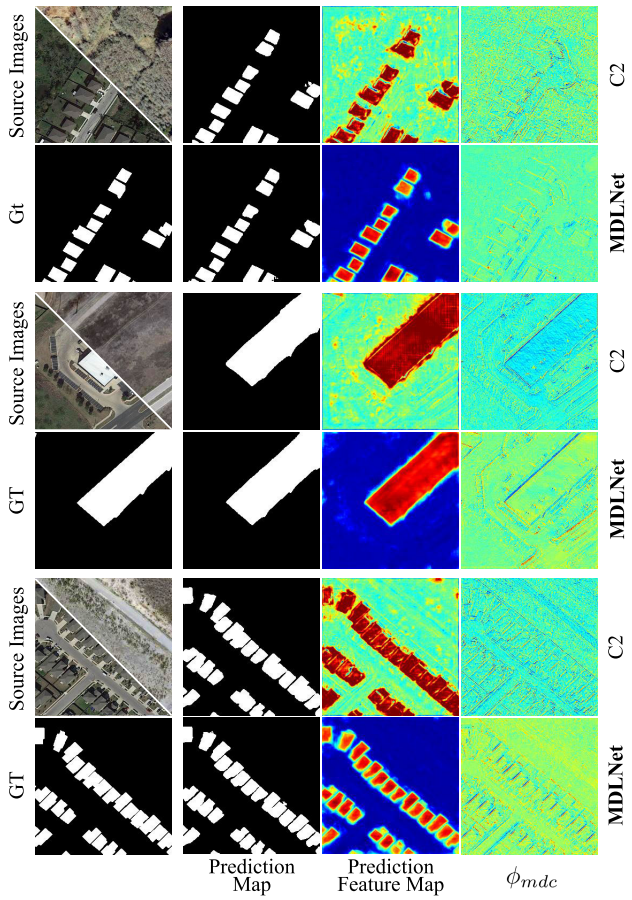


Fig. 5. Visualize feature maps on the LEVIR-CD dataset. C2 is the network that replaces FDIE in MDCM with convolutional operations. The prediction map, prediction feature map, and ϕ_{mdc} of C2 and the proposed MDENet are shown, where ϕ_{mdc} refers to the change feature generated by the first MDCM.

2) *Analysis of MFGatt:* We demonstrate the role of the proposed MFGatt in change detection from two perspectives. First, we directly replace the attention with a 1 × 1 convolution, and the replaced detection result is recorded as C1, which is an intuitive way to prove its effectiveness. Second, when this attention has been removed directly, it only has one 1 × 1 convolution left remaining for the FDIE in MDCM, which undermines the basic feature extraction ability of the MDCM structure. Therefore, in the ablation experiment, we replace this convolution with two 3 × 3 convolutions and mark the replaced detection results as C2. In Table IV, regardless of the replacement methods used, it is evident that the proposed attention can achieve better detection results on the three datasets, especially in the indicators of Rec, F1, IoU, and OA. Specifically, the feature maps generated by C2 and the proposed MDENet are used for qualitative comparison to introduce the effectiveness of the proposed attention and the global feature analysis ability of MDCM. As shown in Fig. 5, the predicted feature maps generated by MDENet have more accurate

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TABLE III
COMPARISON OF DIFFERENT UNITS IN THE MDCM (RED: THE BEST)

Method			LEVIR-CD					WHU-CD					DSIFN-CD				
SDIE_s	SDIE_d	FDIE	Pre	Rec	F1	IoU	OA	Pre	Rec	F1	IoU	OA	Pre	Rec	F1	IoU	OA
✓			90.02	91.05	90.53	82.69	99.04	91.92	94.64	93.26	87.37	99.45	58.85	73.53	65.37	48.56	89.41
✓	✓		89.20	92.13	90.64	82.88	99.06	91.17	92.78	91.97	85.13	99.34	61.95	63.38	62.65	45.62	87.45
✓		✓	89.06	92.22	90.61	82.83	99.06	91.47	94.65	93.04	86.98	99.43	61.32	76.86	68.22	51.76	90.29
✓	✓	✓	89.55	92.68	91.09	83.63	99.11	91.54	95.49	93.47	87.74	99.47	64.61	75.66	69.70	53.49	90.45

TABLE IV
ABLATION EXPERIMENTS OF THE MFGATT (RED: THE BEST)

Method	LEVIR-CD					WHU-CD					DSIFN-CD				
	Pre	Rec	F1	IoU	OA	Pre	Rec	F1	IoU	OA	Pre	Rec	F1	IoU	OA
C1	89.87	91.92	90.88	83.29	99.08	92.54	93.29	92.91	86.77	99.42	70.60	67.57	69.05	52.73	89.25
C2	90.25	91.68	90.96	83.42	99.09	91.59	94.90	93.22	87.29	99.45	74.47	63.34	68.45	52.04	88.34
MDENet	89.55	92.68	91.09	83.63	99.11	91.54	95.49	93.47	87.74	99.47	64.61	75.66	69.70	53.49	90.45

attention to the position of changing objects, while the prediction features generated by C2 are more susceptible to interference from other factors in the background. By observing ϕ_{mdc}, we can find that the proposed method possesses higher edge perception capabilities. Besides, ϕ_{mdc}s of MDENet are smoother than ϕ_{mdc} of C2 in nonedge regions, demonstrating that the proposed method can weaken local noise interference by analyzing global image information. These reflect that MFGatt enhances the image style adaptability of the network to achieve more accurate change detection.

V. CONCLUSION

Toward accurate remote sensing image change detection, we propose a new MDENet. Specifically, an MDCM is designed to simultaneously mine global and local change information, enabling the detection model to have multidimensional foreground and background feature analysis capabilities. In addition, we have also introduced a novel MFGatt to address the interference caused by style differences between images in detection. It purposefully removes style relevance information from images while retaining object position relevance information. The proposed MDENet achieves higher detection accuracy compared with other SOTA change detection methods in experiments conducted on different datasets. However, it involves convolution operations with a high number of channels, leading to a large computational load. To reduce computational costs, we propose a low parameter version of MDENet. It has a similar detection performance as the original version and faster detection efficiency compared with other lightweight methods.

Although the proposed network achieves more advanced performance on multiple remote sensing image change datasets, it still has some potential limitations. Specifically, in large-scale remote sensing image change detection tasks, the proposed method is usually limited by computational environments, such as devices, often necessitating the cropping of source images during the generation of input data. This approach may lead to discontinuities of the detection result. Furthermore, similar to most existing methods, MDENet faces challenges in achieving precise detection across datasets, which affects its practical utility. Therefore, in our future work, we will concentrate on developing detection methods with stronger robustness and lighter weight.

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Jinyang Liu received the B.S. degree from Hunan University, Changsha, China, in 2020, where he is currently pursuing the Ph.D. degree with the Laboratory of Vision and Image Processing.

His research interests include image fusion processing, infrared image processing, and remote sensing image processing.

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Shutao Li (Fellow, IEEE) received the B.S., M.S., and Ph.D. degrees from Hunan University, Changsha, China, in 1995, 1997, and 2001, respectively.

In 2001, he joined the College of Electrical and Information Engineering, Hunan University, where he is currently a Full Professor. From May 2001 to October 2001, he was a Research Associate with the Department of Computer Science, The Hong Kong University of Science and Technology, Hong Kong. From November 2002 to November 2003, he was a Post-Doctoral Fellow with the Royal Holloway College, University of London, London, U.K. From April 2005 to June 2005, he was a Visiting Professor with the Department of Computer Science, The Hong Kong University of Science and Technology. He has authored or coauthored more than 200 refereed articles. His research interests include image processing, pattern recognition, and artificial intelligence.

Dr. Li received the 2nd-Grade State Natural Science Award of China in 2019 and two 2nd-Grade State Scientific and Technological Progress Awards of China in 2004 and 2006. He is an Associate Editor of the IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING and the IEEE TRANSACTIONS ON INSTRUMENTATION AND MEASUREMENT. He is a member of the Editorial Board of *Information Fusion* and *Sensing and Imaging*.



Renwei Dian (Member, IEEE) received the B.S. degree from Wuhan University of Science and Technology, Wuhan, China, in 2015, and the Ph.D. degree from Hunan University, Changsha, China, in 2020.

From November 2017 to November 2018, he was a Visiting Ph.D. Student with the University of Lisbon, Lisbon, Portugal, supported by the China Scholarship Council. From August 2020 to July 2022, he held a post-doctoral position with the College of Electrical and Information Engineering, Hunan University, where he received the distinguished post-doctoral fellowship and is currently an Associate Professor with the School of Robotics. His research interests include hyperspectral image superresolution, image fusion, tensor decomposition, and deep learning.

Dr. Dian was a finalist for the Best Student Paper Award at the International Geoscience and Remote Sensing Symposium (IGARSS) 2018. He received the Wu Wenjun Award for Outstanding Youth in Artificial Intelligence, the Fellowship of China National Postdoctoral Program for Innovative Talents, and an Excellent Doctoral Dissertation by the China Society of Image and Graphics. More information can be found on his homepage https://sites.google.com/view/renweidian/.



Ze Song received the B.S. degree from Hunan University, Changsha, China, in 2020, where he is currently pursuing the Ph.D. degree with the Laboratory of Vision and Image Processing.

His research interests include camouflage scene understanding, multisource image processing, and deep learning.



Xudong Kang (Senior Member, IEEE) received the B.S. degree from Northeast University, Shenyang, China, in 2007, and the Ph.D. degree from Hunan University, Changsha, China, in 2015.

In 2015, he joined the College of Electrical Engineering, Hunan University. His research interests include hyperspectral feature extraction, image classification, image fusion, and anomaly detection.

Dr. Kang received the National Natural Science Award of China (Second Class and Rank as Third) and the Second Prize in the Student Paper Competition in the International Geoscience and Remote Sensing Symposium (IGARSS) 2014. He was selected as the Best Reviewer of the IEEE GEOSCIENCE AND REMOTE SENSING LETTERS and the IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING. He was an Associate Editor of the IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING from 2018 to 2019. He serves as an Associate Editor for the IEEE GEOSCIENCE AND REMOTE SENSING LETTERS and the IEEE JOURNAL ON MINIATURIZATION FOR AIR AND SPACE SYSTEMS.

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